

EXPERIMENT 25

AIM

Combine file metadata and system-log events, normalize different timestamp formats, sort them chronologically and identify important incident events.

ALGORITHM

1. Store evidence events from different sources.
2. Define supported timestamp formats.
3. Parse each timestamp using the available formats.
4. Normalize all timestamps to datetime values.
5. Sort events chronologically.
6. Display normalized timestamp, source, event and object.
7. Use the result as the reconstructed timeline.
8. Stop.

CODE

```
from datetime import datetime

records = [
    ("2026-08-19 09:00:00", "File Metadata", "MODIFIED", "report.docx"),
    ("2026/08/19 09:05:00", "System Log", "LOGIN", "alice"),
    ("19-08-2026 09:10:00", "File Metadata", "CREATED", "suspicious.exe")
]

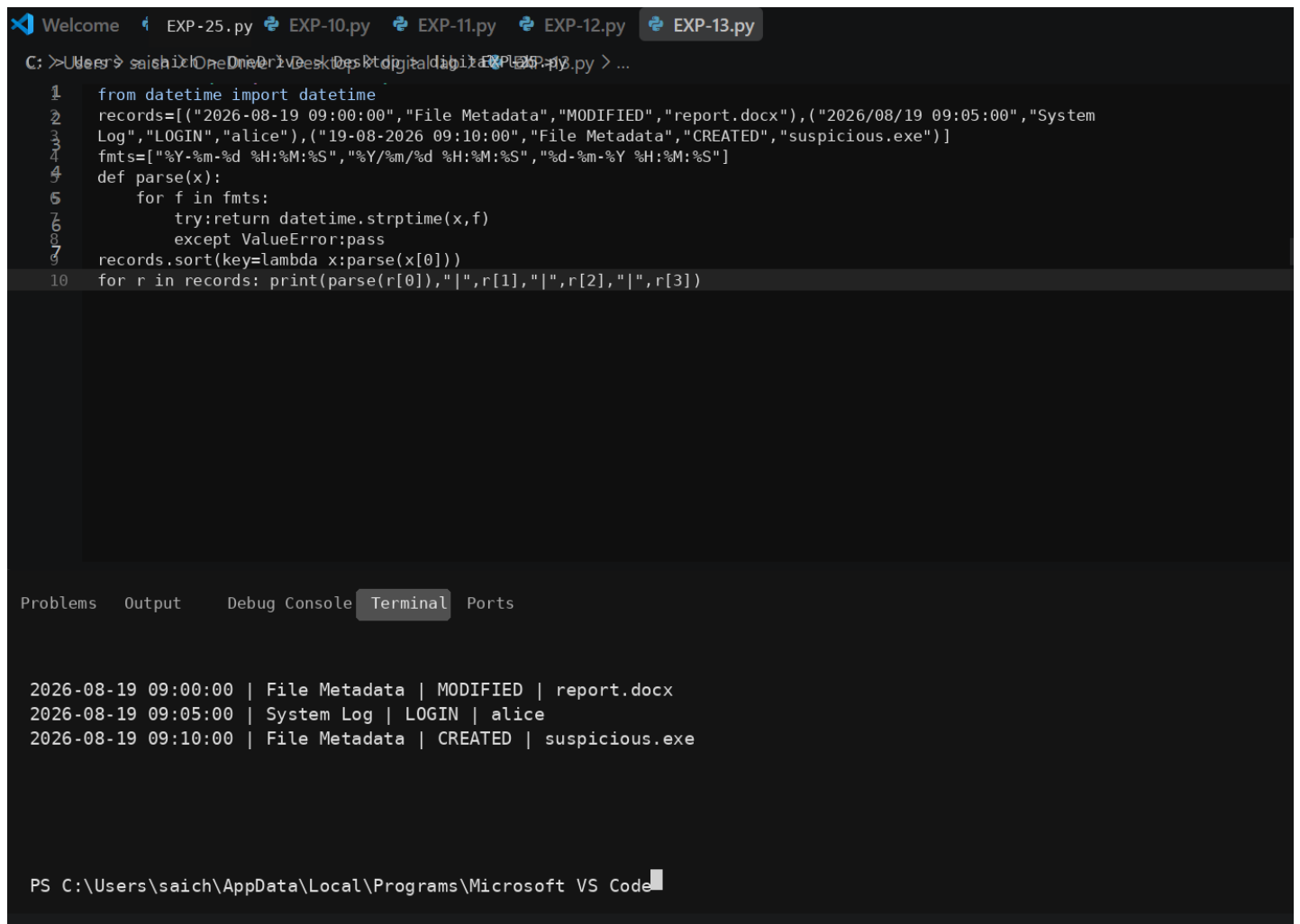
formats = [
    "%Y-%m-%d %H:%M:%S",
    "%Y/%m/%d %H:%M:%S",
    "%d-%m-%Y %H:%M:%S"
]

def parse_timestamp(value):
    for fmt in formats:
        try:
            return datetime.strptime(value, fmt)
        except ValueError:
            pass
    raise ValueError("Unsupported timestamp format")

records.sort(key=lambda item: parse_timestamp(item[0]))

for timestamp, source, event, obj in records:
    print(parse_timestamp(timestamp), "|", source, "|", event, "|", obj)
```

OUTPUT



The screenshot shows a VS Code editor with a file explorer at the top displaying several Python files: EXP-25.py, EXP-10.py, EXP-11.py, EXP-12.py, and EXP-13.py. The main editor window shows the code for EXP-13.py. The code imports the datetime module and defines a list of records. It then defines a parse function that attempts to parse a string into a datetime object using various formats. Finally, it sorts the records by their first element and prints them in a formatted string.

```
1 from datetime import datetime
2 records=[("2026-08-19 09:00:00","File Metadata","MODIFIED","report.docx"),("2026/08/19 09:05:00","System
3 Log","LOGIN","alice"),("19-08-2026 09:10:00","File Metadata","CREATED","suspicious.exe")]
4 fmts=["%Y-%m-%d %H:%M:%S","%Y/%m/%d %H:%M:%S","%d-%m-%Y %H:%M:%S"]
5 def parse(x):
6     for f in fmts:
7         try:return datetime.strptime(x,f)
8         except ValueError:pass
9 records.sort(key=lambda x:parse(x[0]))
10 for r in records: print(parse(r[0]),"|",r[1],"|",r[2],"|",r[3])
```

The terminal output at the bottom shows the results of the script execution:

```
2026-08-19 09:00:00 | File Metadata | MODIFIED | report.docx
2026-08-19 09:05:00 | System Log | LOGIN | alice
2026-08-19 09:10:00 | File Metadata | CREATED | suspicious.exe
```

The terminal prompt shows the current directory as C:\Users\saich\AppData\Local\Programs\Microsoft VS Code.

RESULT

The program reconstructs a single chronological forensic timeline from different evidence sources.